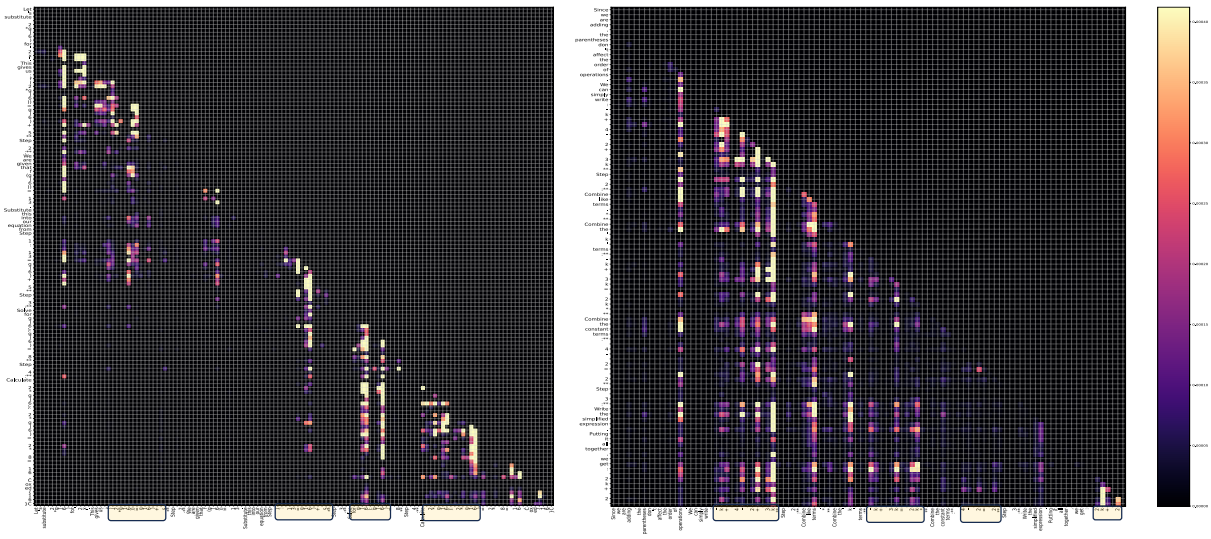

Figure 5: Attention mapping of `DeepSeek-Llama3-8B` and `DeepSeek-qwen-1.5B`.

Here $\alpha_t^{\boldsymbol{x}} = \frac{1}{m}\sum_{i=1}^{m}\alpha_t^{\boldsymbol{r}_i^{\boldsymbol{x}}}$ and $\alpha_t^{\boldsymbol{y}} = \frac{1}{n}\sum_{j=1}^{n}\alpha_t^{\boldsymbol{r}_j^{\boldsymbol{y}}}$ represent the mean activation of all tokens in $\boldsymbol{x}$ and $\boldsymbol{y}$ for feature $\alpha_t$ at layer $L$. Back to our SAE-based steering, to obtain the activation for each feature $\alpha_t$, we computed $\|\alpha_t^{\boldsymbol{x}} - \alpha_t^{\boldsymbol{y}}\|$. It can be derived as follows:

$$\begin{aligned}\boldsymbol{x}-\boldsymbol{y} &= \sum_{t=1}^{d_{\text{SAE}}}\alpha_t^{\boldsymbol{x}}\boldsymbol{z}_t + \boldsymbol{b} + \frac{1}{m}\sum^{m}\boldsymbol{\epsilon} \\ &\quad - \sum_{t=1}^{d_{\text{SAE}}}\alpha_t^{\boldsymbol{y}}\boldsymbol{z}_t - \boldsymbol{b} - \frac{1}{n}\sum^{n}\boldsymbol{\epsilon} \\ &= \sum_{t=1}^{d_{\text{SAE}}}\boldsymbol{z}_t\left(\alpha_t^{\boldsymbol{x}} - \alpha_t^{\boldsymbol{y}}\right) + \left(\frac{1}{m}\sum^{m}\boldsymbol{\epsilon} - \frac{1}{n}\sum^{n}\boldsymbol{\epsilon}\right)\end{aligned} \tag{6}$$

We can approximately assume that the error of each token is equal, so the error term can be removed. If there is an existing SAE, the next step is to sort the activation values $\|\alpha_t^{\boldsymbol{x}} - \alpha_t^{\boldsymbol{y}}\|$ in descending order to get top-k features, then select a certain feature among them to steer the LLM. So we want to find the feature vector $z_t$ such that the above value is as large as possible. More generally, given $N$ pairs of $\{x_p, y_p\}_{p=1}^{N}$, according to equation 2, our goal is to maximze $\sum_{p=1}^{N}\|(\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_p})\|$. To solve this, we need to make following two assumptions.

- *Sparsity Assumption*. Since the features activated by SAE are sparse, we generally only look for the top $K$ significant features, like $K = 10, 100, 500$, so we prefer the activation values to be distributed on these $K$ features, which again enhances the sparsity of SAE. To verify this assumption, we quantitatively analyzed the relationship between the percentage of restored norm after reconstruction and the number of features required for reconstruction in Figure 6. We can find that the norm after reconstruction using the top-10 features has recovered more than 90%. When $K = d_{model}$, the reconstruction has been restored to around 99%. It means that utilizing only the top-k features to reconstruct the original vector does not lose much information.So we can assume that $\sum_{t=1}^{d_{\text{SAE}}}\boldsymbol{z}_t(\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_p}) \approx \sum_{t=1}^{\hat{d}}\boldsymbol{z}_t\left(\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_p}\right)$, $\hat{d} < d_{model} \ll d_{SAE}$.
- *Orthogonality Assumption.* As evidenced by Figure 7, the feature vectors $\boldsymbol{z}_t$ associated with the top-k features $\alpha_t$ exhibit approximate pairwise orthogonality. So we can assume that we are looking for $\hat{d}$ mutually orthogonal $\boldsymbol{z}_t$.

**Solution for Steering vector.** Based on above assumptions, we have $\|\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_k}\| = \|\langle \boldsymbol{x}_p - \boldsymbol{y}_p, \boldsymbol{z}_t\rangle\|$ by Appendix E. So if we let $\boldsymbol{A} = (\boldsymbol{x}_1 - \boldsymbol{y}_1, \boldsymbol{x}_2 - \boldsymbol{y}_2, ..., \boldsymbol{x}_N - \boldsymbol{y}_N)$, it turns out to be the following problem:

$$\begin{aligned}&\max_{\|\boldsymbol{z}_t\|=1}\sum_{p=1}^{N}\|\alpha_t^{\boldsymbol{x}_p} - \alpha_t^{\boldsymbol{y}_p}\| \\ &= \max_{\|\boldsymbol{z}_t\|=1}\sum_{p=1}^{N}\|\langle \boldsymbol{x}_p - \boldsymbol{y}_p, \boldsymbol{z}_t\rangle\| \\ &\propto \max_{\|\boldsymbol{z}_t\|=1}\|\boldsymbol{A}^\top\boldsymbol{z}_t\|^2 = \max_{\|\boldsymbol{z}_t\|=1}\boldsymbol{z}_t^\top\boldsymbol{A}\boldsymbol{A}^\top\boldsymbol{z}_t\end{aligned} \tag{7}$$

The solving process can be found in Appendix E.

The goal is to get the unit eigenvectors $\{\boldsymbol{u}_t\}_{t=1}^{d_{model}}$ of $\boldsymbol{A}\boldsymbol{A}^T$ as descending order of eigenvalues, then select a certain eigenvector $\boldsymbol{u}_t$ to steer the residual activations $\boldsymbol{r}^h$ as follows:

$$\hat{\boldsymbol{r}}_i^h = \boldsymbol{r}_i^h + \lambda \boldsymbol{u}_t, i \in \mathcal{H} \tag{8}$$

## 5 Experiments of SAE-free Steering

### 5.1 Implementation Details

**Experiment Setting.** We leverage `DeepSeek-Llama3-8B` and `DeepSeek-qwen-1.5B` as our evaluation LLMs. There is no corresponding pretrained SAE for `DeepSeek-qwen-1.5B`. We choose hidden states in layer 15 to obtain $\boldsymbol{r}_i^x$ and $\boldsymbol{r}_j^y$. The setting of the dataset and generation remains identical to that described in the preceding section.

**Baselines.** We use BoostStep (Zhang et al., 2025) and MathNeuro (Christ et al., 2024) as comparing baselines. BoosStep decomposes the process of solving problems into many steps. When generating each step, the method will try to retrieve the most similar sub-steps from a reasoning steps bank. The goal of MathNeuro is to find the math-related neurons in the LLM and then scale their activation.

### 5.2 Comparing with Baselines

The result is shown in Table 2. It indicates that compared with the baseline, SAE-free steering has a consistent improvement on each dataset. For MATH-L3&L4 and MMLU-high, SAE-free steering outperforms SAE-based steering. The results demonstrate that the information extracted from SAE-free steering exhibits superior applicability. The enhanced performance may be attributed to the fact that SAEs are trained on massive datasets, potentially introducing noise into their feature representations. In contrast, our proposed SAE-free approach derives its information directly from structured reasoning processes, yielding more targeted and reliable features.

### 5.3 Attention Analysis

We also visualize the effect of SAE-free steering from the perspective of attention distribution. We calculate the change of attention distribution before and after the intervention in the next layer of the steering layer. As is shown in Figure 5, the left and right are the visualizations of `DeepSeek-qwen-1.5B` and `DeepSeek-Llama3-8B`, the indexes in the x-axis with yellow boxes surrounded indicate tokens representing mathematical representations. We can observe that the model will pay more attention to these tokens after steering, which aligns well with our expectations. It means that steering activations with these eigenvalues will lead to more reasonable attention allocation.

## 6 Related Work

**Sparse Autoenocder.** SAE is trained to reconstruct the middle activations (Ng et al., 2011; Karvonen et al., 2024; Marks et al., 2024). Various SAE variants have been proposed to address the limitations of traditional SAEs, such as Gated SAE (Rajamanoharan et al., 2024a), TopK-SAE (Gao et al., 2024), and JumpReLU SAE (Rajamanoharan et al., 2024b). SAE has been proven useful in many fields of application in LLM, such as knowledge editing (Zhao et al., 2024), machine unlearning (Muhamed et al., 2025; Farrell et al., 2024), improving safety (Wu et al., 2025b), and robustness (Wu et al., 2025a).

**Representation Engineering.** Representation engineering has emerged as an effective approach to enhance the interpretability and transparency of LLMs. Zou et al. (2023) has summarized recent progress and applications of representation engineering in bias, fairness, model editing, and other fields. Li et al. (2024) has utilized this technique to quantify LLMs' performance in different languages. Li et al. (2023) reveals that the internal representations of attention heads within LLMs serve as reliable indicators of reasoning paths. Through probe-based interventions, they correct the internal representation direction to improve the LLM's performance.

## 7 Conclusions

In this paper, we introduce a novel framework for enhancing reasoning abilities in LLMs through targeted feature extraction and steering of residual activations. By decomposing vanilla CoT reasoning into verbal and symbolic processes, we identified and amplified task-relevant reasoning features while suppressing noise, demonstrating consistent improvements across multiple mathematical reasoning benchmarks. Additionally, our SAE-free steering algorithm extends these benefits to models without corresponding pre-trained SAEs by computing steering directions directly from residual activations, achieving comparable or even superior results to SAE-based methods in some scenarios.

## Limitations

While our feature extraction and steering methods demonstrate promising results in enhancing chain-of-thought reasoning capabilities, there are several limitations. First, our evaluation focuses mainly on mathematical reasoning datasets (GSM8K, MATH-L3&L4, MMLU-high, and MathOAI). In future, we plan to extend our proposed steering techniques to more diverse reasoning domains such as logical reasoning, scientific problem-solving, or multi-step planning tasks that don't primarily involve mathematics. Second, our current experiments are limited to three model families (LLaMA, DeepSeek, and Gemma). We plan to apply our method to more diverse LLM architectures to verify the generalizability of our approach.

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